PROBABILITY & STATISTICS (SEPS-204)

I. Course Details

Credit Hours:	3 (3+0)
Pre-requisites:	Calculus
Course Leader	Dr. Asad Husain
Recommended Textbook(s)	Probability and Statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying E. Ye Publisher Pearson.
Recommended Reference (Books/Websites/Articles)	1. Introduction to Statistical Theory by Sher Muhammad Chaudhry, Dr. Shahid Kamal, Ninth Edition 2013, 2. Probability and Statistics for Engineers and Scientists by Anthony J. Hayter. 3. Schaum's Outline of Probability and Statistics, by John Schiller, R. Alu Srinivasan and Murray Spiegel, McGraw-Hill; 3rd Edition (2008). ISBN-10:0071544259

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	**Explain** fundamental concepts of probability and statistics along with graphical representation to state their main ideas.	Cognitive	2	1	A1, Q1, Mid Term
CLO-2	**Apply** probability formulas and distributions to solve real-world problems.	Cognitive	3	2	A2, Q2, Mid Term, Final Term
CLO-3	**Apply** basic statistical techniques to engineering data for solving various problems.	Cognitive	3	2	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	**Weight (%)**
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	**100**

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to Statistics and Data Analysis, Statistical Inference, Samples, Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Probability: Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes' Rule. Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem. Sampling Distribution of S2, t-Distribution. Tests of Hypotheses. The Use of P Values for Decision Making in Testing Hypotheses (Single Sample & One- and Two Sample Tests), Linear Regression and Correlation. Least Squares and the Fitted Model, Multiple Linear Regression and Certain, Nonlinear Regression Models, Linear Regression Model Using Matrices, Properties of the Least Squares Estimators.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to Statistics and Data Analysis: Descriptive and inferential statistics, population and sample, Observations and variables, Types of variables, Data collection	Chapter 1
2		Measure of central tendency and Measure of dispersion, quartiles for ungroup data, Graphical representation of data: dot plot, Stem leaf Display, Box and whisker plot	Chapter 1
3	CLO-2	Introduction to Probability, concept of sets, Venn diagram, operation and algebra on sets, Cartesian product	Chapter 2
4		Counting Sample Points, Sample Space, Events	Chapter 2
5		Definition of probability axioms of probability, conditional probability, independent events,	Chapter 2

6		Additive Rules and the Product Rule, Bayes' Rule	Chapter 2
7		Random variables, Mathematical Expectation: Mean of a Random Variable, Variance of Random Variables,	Chapter 3,4
8		Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem	Chapter 3,4
9	**Midterm Exams**		
10	CLO-2	Discrete Probability Distributions: Binomial distribution, Poisson distribution	Chapter 5
11		Continuous Probability Distributions: Normal distribution and standard normal distribution and its properties	Chapter 6
12		Random Sampling, Sampling Distribution of Means and the Central Limit Theorem	Chapter 8
13	CLO-3	Linear Regression Model using Matrices, Least Squares and the Fitted Model	Chapter [Ref. book1]
14		Multiple Linear Regression, coefficient of correlation	Chapter [Ref. book1]
15		Tests of Hypotheses: The Use of P Values for Decision Making in Testing Hypotheses, alpha level, significance	Chapter [Ref. book1]
16		Sampling Distribution of S2, Chi-squared distribution, t-Distribution	Chapter [Ref. book1]